

VELTRAXX'26



इलेक्ट्रॉनिकी एवं
सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
ELECTRONICS AND
INFORMATION TECHNOLOGY



Chips to Startup
Programme



INSTITUTION'S
INNOVATION
COUNCIL
(Ministry of Education Initiative)

BAJAJ INSTITUTE OF TECHNOLOGY, WARDHA

TEAM NAME: SILICON SYNDICATE

PROBLEM STATEMENT NO: PS04

PROBLEM STATEMENT TITLE: Configurable Dual-Clock Asynchronous FIFO with Gray Code Pointers.

PS Category- RTL-to-GDSII

VELTRAXX'26



Problem Statement

Configurable Dual-Clock Asynchronous FIFO with Gray Code Pointers

Design a highly robust, parameterizable asynchronous FIFO (configurable depth N and data width W) to guarantee reliable data transfer between two independent, non-integer-multiple clock domains. The architecture must successfully navigate Clock Domain Crossing (CDC) hazards, ensuring absolute data integrity and glitch-free control signaling despite unrelated clock frequencies and dynamic phase shifts.

VELTRAXX'26



YOUR SOLUTION

PROJECT NAME : CONFIGURABLE DUAL CLOCK ASYNCHRONOUS FIFO

ONE-LINE SOLUTION : A dual clock FIFO using Gray coded pointers and 2 stage synchronizers to safely transfer data between independent clock domains

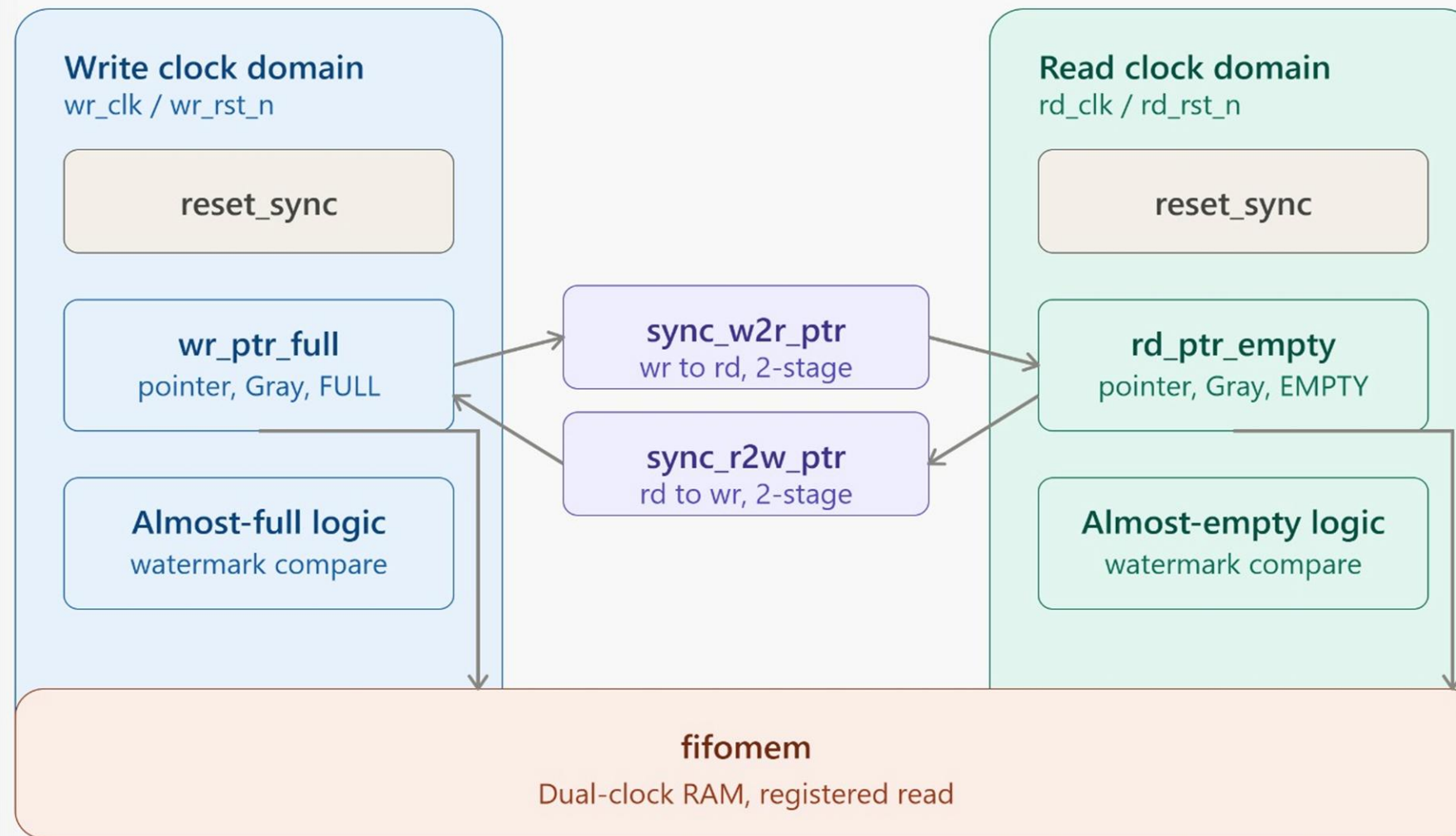
HOW IT SOLVES THE PROBLEM:

- Dual-clock operation enables safe data transfer between independent write and read domains.
- Gray-coded read/write pointers minimize CDC errors by changing only one bit at a time.
- 2-stage synchronizers safely transfer pointer information across clock domains and reduce Metastability risk.
- FULL and EMPTY logic prevents FIFO overflow and underflow during asynchronous operation.

Almost-FULL and Almost-EMPTY flags provide early status indication for better flow control.

VELTRAXX'26

ARCHITECTURAL BLOCK DIAGRAM



VELTRAXX'26



PURPOSE OF BLOCKS

BLOCK	PURPOSE	WHY CHOSEN
FIFO Memory	Stores transferred data	Separates write/read timing
Write Pointer	Tracks write location	Independent write-domain control
Read Pointer	Tracks read location	Independent read-domain control
Gray Converter	Converts binary pointer to Gray	Only one bit changes between adjacent states
2-FF Synchronizer	CDC pointer transfer	Reduces metastability propagation
FULL Logic	Prevents overflow	Blocks writes when capacity is reached
EMPTY Logic	Prevents underflow	Blocks reads when no data is available
Almost Flags	Early warning	Allows upstream/downstream logic to react

• Advantages

- Robust CDC
- Independent clock frequencies
- No data loss from clock mismatch
- Simple scalable architecture

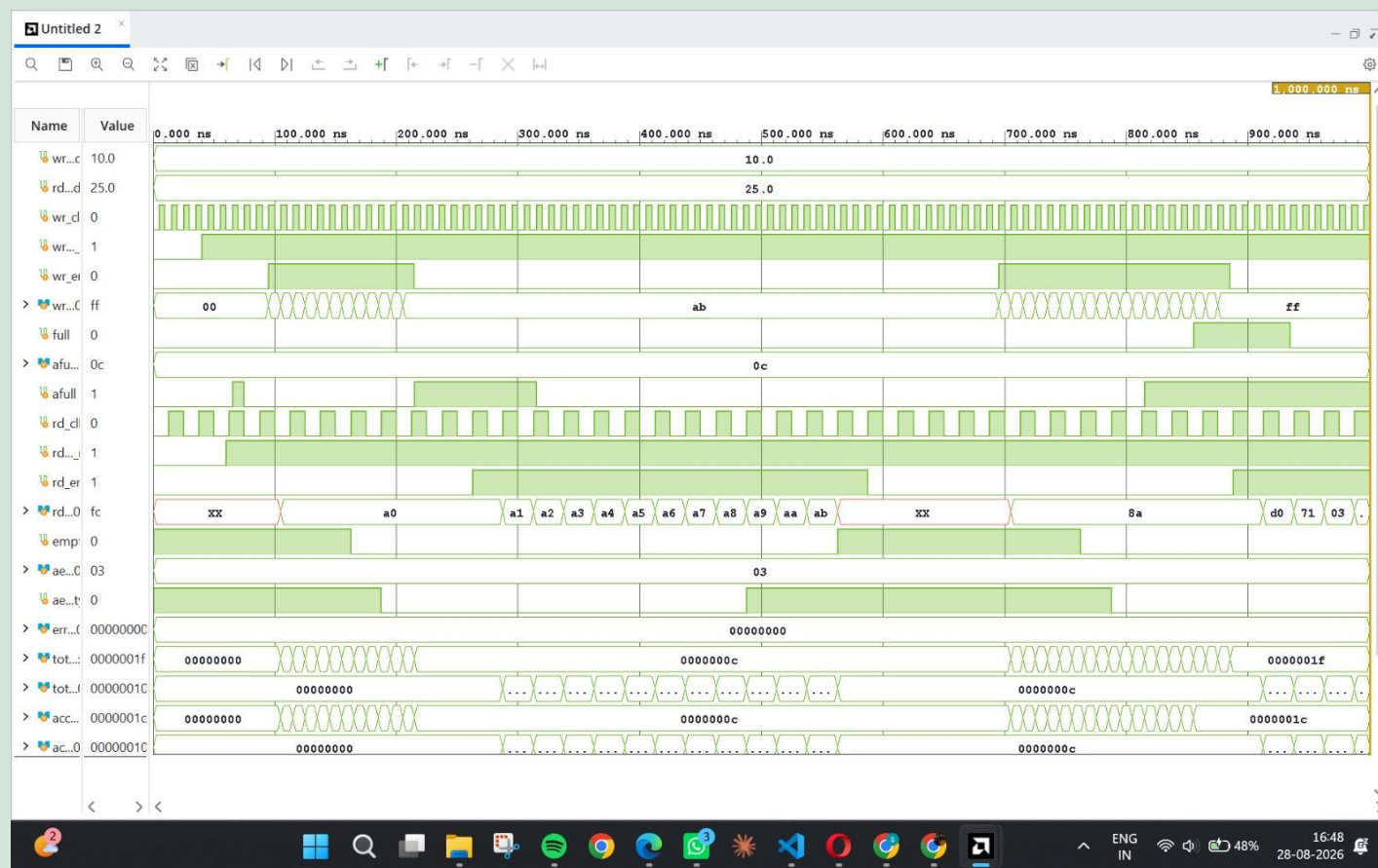
• Trade-offs

- 2-cycle synchronization latency
- Additional pointer/synchronizer registers
- More logic than a single-clock FIFO



VELTRAXX'26

OUTPUTS AND RESULTS

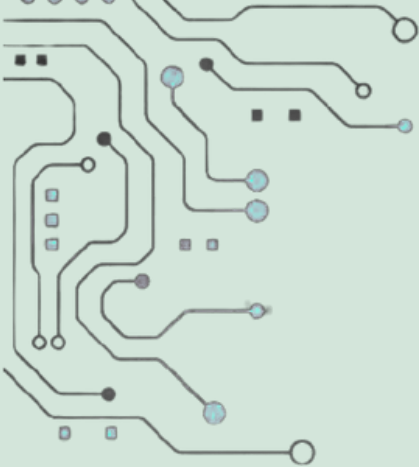


Simulation waveforms

```
1 + analyze_power_grid -net VDD -voltage_file /home/jayanth/librelane/designs/asyncfifo/runs/RUN_2026-08-28_15-48-26/57-openroad-irdropreport/net-VDD.csv
2 [INFO PSM-0040] All shapes on net VDD are connected.
3 ##### IR report #####
4 Net      : VDD
5 Corner   : nom_tt_025C_1v80
6 Total power   : 4.11e-03 W
7 Supply voltage : 1.80e+00 V
8 Worstcase voltage: 1.80e+00 V
9 Average voltage : 1.80e+00 V
10 Average IR drop : 3.70e-05 V
11 Worstcase IR drop: 1.09e-04 V
12 Percentage drop : 0.01 %
13 #####
14 + analyze_power_grid -net VSS -voltage_file /home/jayanth/librelane/designs/asyncfifo/runs/RUN_2026-08-28_15-48-26/57-openroad-irdropreport/net-VSS.csv
15 [INFO PSM-0040] All shapes on net VSS are connected.
16 ##### IR report #####
17 Net      : VSS
18 Corner   : nom_tt_025C_1v80
19 Total power   : 4.11e-03 W
20 Supply voltage : 0.00e+00 V
21 Worstcase voltage: 1.22e-04 V
22 Average voltage : 3.62e-05 V
23 Average IR drop : 3.62e-05 V
24 Worstcase IR drop: 1.22e-04 V
25 Percentage drop : 0.01 %
26 #####
```

IR drop report





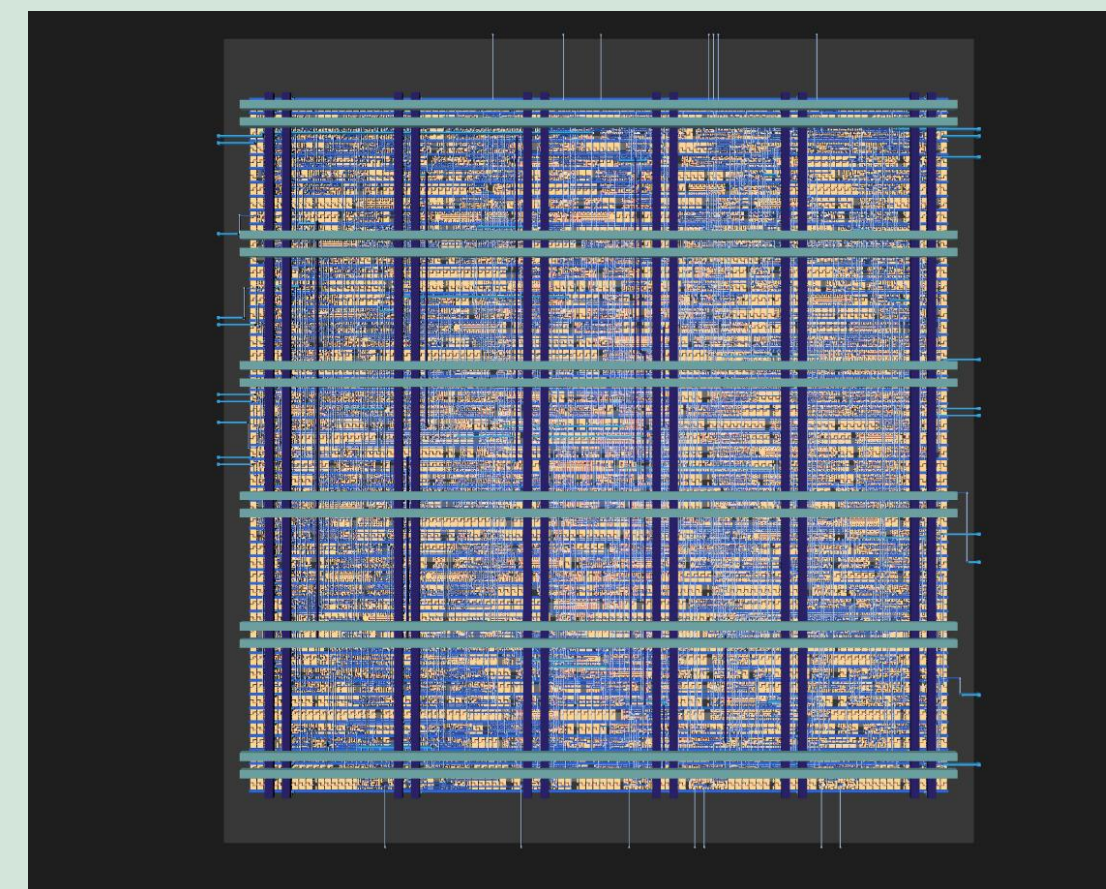
VELTRAXX'26



OUTPUTS AND RESULTS

Corner/Group	Hold Worst Slack	Reg to Reg Paths	Hold TNS	Count	of which reg to reg	Setup Worst Slack	Reg to Reg Paths	Setup TNS	Setup Vio Count	of which reg to reg	Max Cap Violatio...	Max Slew Violati...
Overall	0.1237	0.1237	0.0000	0	0	2.9926	2.9926	0.0000	0	0	0	80
nom_tt_025C_1v80	0.2598	0.2598	0.0000	0	0	6.3445	6.3445	0.0000	0	0	0	0
nom_ss_100C_1v60	0.6154	0.6154	0.0000	0	0	3.0854	3.0854	0.0000	0	0	0	54
nom_ff_n40C_1v95	0.1240	0.1240	0.0000	0	0	7.0780	8.6351	0.0000	0	0	0	0
min_tt_025C_1v80	0.2594	0.2594	0.0000	0	0	6.3892	6.3892	0.0000	0	0	0	0
min_ss_100C_1v60	0.6046	0.6046	0.0000	0	0	3.1633	3.1633	0.0000	0	0	0	36
min_ff_n40C_1v95	0.1237	0.1237	0.0000	0	0	7.1075	8.6555	0.0000	0	0	0	0
max_tt_025C_1v80	0.2604	0.2604	0.0000	0	0	6.2936	6.2936	0.0000	0	0	0	0
max_ss_100C_1v60	0.6238	0.6238	0.0000	0	0	2.9926	2.9926	0.0000	0	0	0	80
max_ff_n40C_1v95	0.1245	0.1245	0.0000	0	0	7.0473	8.6156	0.0000	0	0	0	0

Timing Report



GDS View



VELTRAXX'26



RESEARCH AND REFERENCES

- **Design, Simulation and Realization of a Parametrizable, Configurable and Modular Asynchronous FIFO**

By: Haytham Ashour Mentor Graphics, Emulation Division Cairo, Egypt haytham_ashour@mentor.com

